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2026-06-17

High tolerance of microzooplankton to ocean acidification in an Arctic coastal plankton community

2013

Aberle, N.; Schulz, K. G.; Stuhr, A.; Malzahn, A. M.; Ludwig, A.; Riebesell, U.

[Paper URL](#)

Abstract: Impacts of ocean acidification (OA) on marine biota have been observed in a wide range of marine systems. We used a mesocosm approach to study the response of a high Arctic coastal microzooplankton community during the post-bloom period in Kongsfjorden (Svalbard) to direct and indirect effects of high pCO₂/low pH. We found almost no direct effects of OA on microzooplankton composition and diversity. Both the relative shares of ciliates and heterotrophic dinoflagellates as well as the taxonomic composition of microzooplankton remained unaffected by changes in pCO₂/pH. Although the different pCO₂ treatments affected food availability and phytoplankton composition, no indirect effects (e.g. on the total carrying capacity and phenology of microzooplankton) could be observed. Our data point to a high tolerance of this Arctic microzooplankton community to changes in pCO₂/pH. Future studies on the impact of OA on plankton communities should include microzooplankton in order to test whether the observed low sensitivity to OA is typical for coastal communities where changes in seawater pH occur frequently.

TAGS

general

m_cdr_focus: associated_fields | **m_cdr_method:** not_applicable | **paper_type:** original_research | **paper_topic:** community_impacts; species_sensitivity | **type_of_method_used:** mesocosm | **adjacent_topic_to_m_cdr:** ocean_acidification | **adjacent_topic_to_fisheries:** lower_trophic

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 7/7/2025

location

ocean_basin: arctic | **geopolitical_area:** europe | **habitat_type:** nearshore | **depth:** surface | **experiment_location:** mesocosm

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** zooplankton | **life_stage:** not_applicable

treatment

exposure: chemical | **chemical_mineral_added:** other | **response_observed:** no_effect_on_biology

NOTES

summary_notes: OA study looking at high pCO₂/low pH effects on zooplankton community.

mcd_r_relevance_notes: Biological response to carbonate chemistry change (pH).

fisheries_relevance_notes: Microzooplankton composition and diversity was not directly or indirectly affected, suggesting tolerance to pH change.

Dissecting the impact of CO₂ and pH on the mechanisms of photosynthesis and calcification in the coccolithophore *Emiliana huxleyi*

2013

Bach, Lennart T.; Mackinder, Luke C. M.; Schulz, Kai G.; Wheeler, Glen; Schroeder, Declan C.; Brownlee, Colin; Riebesell, Ulf

[Paper URL](#)

Abstract: Coccolithophores are important calcifying phytoplankton predicted to be impacted by changes in ocean carbonate chemistry caused by the absorption of anthropogenic CO₂. However, it is difficult to disentangle the effects of the simultaneously changing carbonate system parameters (CO₂, bicarbonate, carbonate and protons) on the physiological responses to elevated CO₂. Here, we adopted a multifactorial approach at constant pH or CO₂ whilst varying dissolved inorganic carbon (DIC) to determine physiological and transcriptional responses to individual carbonate system parameters. We show that *Emiliana huxleyi* is sensitive to low CO₂ (growth and photosynthesis) and low bicarbonate (calcification) as well as low pH beyond a limited tolerance range, but is much less sensitive to elevated CO₂ and bicarbonate. Multiple up-regulated genes at low DIC bear the hallmarks of a carbon-concentrating mechanism (CCM) that is responsive to CO₂ and bicarbonate but not to pH. *Emiliana huxleyi* appears to have evolved mechanisms to respond to limiting rather than elevated CO₂. Calcification does not function as a CCM, but is inhibited at low DIC to allow the redistribution of DIC from calcification to photosynthesis. The presented data provides a significant step

in understanding how *E. huxleyi* will respond to changing carbonate chemistry at a cellular level.

TAGS

general

m_cdr_focus: associated_fields | **m_cdr_method:** not_applicable | **paper_type:** original_research | **paper_topic:** species_sensitivity | **type_of_method_used:** lab_experiment | **adjacent_topic_to_m_cdr:** fundamental_carbon_chem; ocean_acidification | **adjacent_topic_to_fisheries:** lower_trophic

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 7/9/2025

location

ocean_basin: not_applicable | **geopolitical_area:** not_applicable | **habitat_type:** not_applicable | **depth:** not_applicable | **experiment_location:** laboratory

species

species_common_name: NA | **species_scientific_name:** *Emiliana huxleyi* | **taxon:** phytoplankton | **life_stage:** not_applicable

treatment

exposure: chemical; low_co2 | **chemical_mineral_added:** other | **response_observed:** biological_effect_observed

NOTES

summary_notes: Experimental study of the sensitivity of *Emiliana huxleyi* (coccolithophore) to changes in carbonate system parameters (CO₂, bicarbonate, carbonate, pH).

mcdr_relevance_notes: Carbonate chemistry changes induced by mCDR may impact calcification and photosynthesis by *E. huxleyi*.

fisheries_relevance_notes: *Emiliana huxleyi* is sensitive to low CO₂ (growth and photosynthesis) and low bicarbonate (calcification), while less sensitive to elevated bicarbonate.

Final Report of the Shipboard Testing of the Sodium Hydroxide (NaOH) Ballast Water Treatment System Onboard the MV Indiana Harbor

2013

Cangelosi, Allegra

[Paper URL](#)

Abstract: In the summer of 2010, the National Parks of Lake Superior Foundation and researchers from the U.S. Geological Survey’s Leetown Science Center (USGS), received support from the USEPA’s Great Lakes Restoration Initiative (GLRI) to develop and trial a full-scale BWTS involving NaOH with applicability to U.S. flag vessels in Great Lakes trade. As part of this project, the research team enlisted GSI to undertake a status test on BWTS’ biological effectiveness and residual toxicity in the context of a single shipboard trial (one ballast uptake operation, one retention period, and one ballast discharge operation). The installation to be tested was a temporary and partial (two tank) prototype installed in two tanks on board the motor vessel (MV) Indiana Harbor, with alternate dosing approaches in each of the two tanks. The subject BWTS involved elevating pH by adding sodium hydroxide (NaOH, in the same formulation used for lye or caustic soda), retaining treated ballast water for a minimum period, and then neutralizing the ballast water prior to discharge. GSI’s status test involved collecting preliminary data on the biological treatment efficacy and residual toxicity (i.e., via whole effluent toxicity, WET, testing) from a single demonstration voyage based on measurement of ballast uptake into and discharge from two treatment tanks and two control tanks. GSI developed a detailed test plan that described the design of the single biological efficacy trial (including sample collection, analysis endpoints, sample handling and custody, WET, and data collection and recording), which was subject to review and comment by the NaOH BWTS development team prior to finalization (GSI, 2011). The GSI status test began on August 18, 2011, during normal vessel ballast intake operations in the port of Gary, Indiana, and concluded three days later on August 22 during normal vessel ballast discharge operations in the port of Superior, Wisconsin. On intake, GSI sampled harbor water that was loaded into four of the ship’s port side tanks (2P, 3P, 4P and 5P). There were adequate numbers of live zooplankton in the intake water (i.e., 43,000/m³ to 235,000/m³ of live organisms 50 μ m) to warrant continuation of the trial. The water in two of these tanks (3P and 4P) was concurrently dosed with enough 50 % (w/v) NaOH solution to achieve a pH of about 12. Approximately 18 hours prior to the MV Indiana Harbor’s arrival in Superior an in-tank carbonation system was activated in both treatment tanks to neutralize the pH of the treated water to below 8.8, i.e., a level considered safe for release into the receiving harbor. Following the vessel’s arrival in port, the ballast water from the treatment tanks and the untreated water from the control tanks was discharged in sequence and sampled. As a single replicate, this GSI status test of the prototype BWTS is in no way conclusive or determinative. The results reported here provide only an indication of the system’s potential effectiveness relative to no treatment. In this single trial, BWTS-treated discharge contained live organisms 50 μ m (i.e., zooplankton) in concentrations ranging 178/m³ to 441/m³. These concentrations are lower than control discharge densities which ranged from 100,000/m³ to 167,000/m³. Densities of live organisms 10 and < 50 μ m in the treatment discharge ranged from 2 cell/mL to 8 cells/mL, while control discharge concentrations were higher, ranging from 53 cells/mL to 92 cells/mL. In terms of organisms < 10 μ m, the trial produced inconclusive results with concentrations of both total coliforms and heterotrophic bacteria highest in discharge samples from one of the treatment tanks (4P). The results from a WET test indicate that the treated and neutralized discharge water produced no residual toxicity to green algae (*Selenastrum capricornutum*) or the fathead minnow (*Pimephales promelas*). However, in these tests, the treated ballast water

significantly affected both survival and reproduction of the cladoceran *Ceriodaphnia dubia*, indicating possible residual toxicity. The BWTS developer asserts that this toxicity could derive from artifactual pH drift during the WET test; pH increased by a maximum of about one unit over the 24 hour period following each daily renewal (Appendix 1). The GSI team did not control pH drift in daily exposures during the WET tests to avoid altering the inherent properties (including conductivity) of the discharge water subject to toxicity testing. Overall, the BWTS warrants further development and evaluation at the land- and ship-based levels.

TAGS

general

m_cdr_focus: associated_fields | **m_cdr_method:** not_applicable | **paper_type:** report | **paper_topic:** biological_ecological_impacts; ecotoxicity; tech_engineering | **type_of_method_used:** field_study | **adjacent_topic_to_m_cdr:** freshwater_environments; other | **adjacent_topic_to_fisheries:** other

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 7/29/2025

location

ocean_basin: not_applicable | **geopolitical_area:** united_states | **habitat_type:** not_applicable | **depth:** not_applicable | **experiment_location:** field

species

species_common_name: NA | **species_scientific_name:** *Selenastrum capricornutum*; *Pimephales promelas*; *Ceriodaphnia dubia* | **taxon:** crustacean; fish; phytoplankton | **life_stage:** not_applicable

treatment

exposure: chemical | **chemical_mineral_added:** naoh | **response_observed:** biological_effect_observed

NOTES

summary_notes: Report on ballast water treatment system (designed to prevent release of invasive species) using NaOH to adjust pH up to ~12; water was then neutralized before release at pH <8.8. The residual toxicity findings may be relevant to mCDR (discharge of water with elevated pH relative to ambient harbor conditions).

mcd_r_relevance_notes: Residual toxicity from discharge of treated water, pH adjusted to below 8.8.

fisheries_relevance_notes: Treated and neutralized discharge water wasn't toxic to green algae (*Selenastrum capricornutum*) or the fathead minnow (*Pimephales promelas*), but did impact the survival and reproduction of the cladoceran *Ceriodaphnia dubia*.

Development of air-lifted artificial upwelling powered by wave

2013

Chen, Jiawang; Yang, Jing; Lin, Shan; Fan, Wei; Chen, Ying; Liang, Naikuang; Ge, Han; Huang, Haocai

[Paper URL](#)

Abstract: This paper presents that an artificial upwelling system integrated with air-lift upwelling tube and wave powered air booster. The upwelling flow had proved that the air-water mixture flow upward in upwelling tube immersed water can make the underwater at bottom flow upward too. This paper introduces the wave power to collect and pump air into water in one depth of upwelling tube. Meanwhile in the deep water far from the continent, the nutrition concentration near the seabed is higher than that near the surface of sea. On the contrary, the oxygen concentration near the seabed is less than that on the top. The upwelling flow is an available method to balance or decrease the difference of characters between surface and bottom water in the sea, which can improve the environment of marine ranching, and promote the growth of the bottom of the biological food chain. One project - air-lifted artificial upwelling powered by wave had been applied and carried on. In order to confirm the air-lifted upwelling efficiency, 2 lake experiments were finished in Qiandao Lake, The first experiment was done directly with a plastic hose, while at the second experiment, the upwelling tube made of tensional cloth is supported by steel supporting rings, these experiments showed that the efficiency of upwelling flow of the plastic hose is higher than that of upwelling tube with supporting rings, an upwelling tube shrinkage deformation happens easily in the case of large upwelling flow. In addition, a simulation platform was designed and numerical simulated, which simulate the wave lifting and dropping the buoyant ball then get compressed air, whose aim is to get the performance of air collecting and compressing capacity and the efficiency between the hydraulic power and compressed air power.

TAGS

general

m_cdr_focus: associated_fields | **m_cdr_method:** not_applicable | **paper_type:** original_research | **paper_topic:** tech_engineering | **type_of_method_used:** field_study; modeling | **adjacent_topic_to_m_cdr:** other | **adjacent_topic_to_fisheries:** not_applicable

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 7/30/2025

location

ocean_basin: not_applicable | **geopolitical_area:** not_applicable | **habitat_type:** not_applicable | **depth:** not_applicable | **experiment_location:** field

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** not_applicable
| **life_stage:** not_applicable

treatment

exposure: not_applicable | **chemical_mineral_added:** not_applicable | **response_observed:** biological_effect_not_investigated

NOTES

summary_notes: Proposed design of artificial upwelling system using wave power. Two tests were performed in lakes to demonstrate efficiency.

mcd_r_relevance_notes: Engineering design for system that could be applied for upwelling.

fisheries_relevance_notes: Not relevant

Installing kelp forests/seaweed beds for mitigation and adaptation against global warming: Korean Project Overview

2013

Chung, Ik Kyo; Oak, Jung Hyun; Lee, Jin Ae; Shin, Jong Ahm; Kim, Jong Gyu; Park, Kwang-Seok

[Paper URL](#)

Abstract: Chung, I. K., Oak, J. H., Lee, J. A., Shin, J. A., Kim, J. G., and Park, K.-S. 2013. Installing kelp forests/seaweed beds for mitigation and adaptation against global warming: Korean Project Overview. – ICES Journal of Marine Science, 70: 1038–1044. Seaweed beds can serve as a significant carbon dioxide (CO₂) sink while also satisfying global needs for food, fodder, fuel, and pharmaceutical products. The goal of our Korean Project has been to develop new baseline and monitoring methodologies for mitigation and adaptation within the context of climate change. Using innovative research approaches, we have established the Coastal CO₂ Removal Belt (CCRB), which comprises both natural and man-made plant communities in the coastal region of southern Korea. Implemented on various spatial–temporal scales, this scheme promotes the removal of CO₂ via marine forests. For example, when populated with the perennial brown alga *Ecklonia*, a pilot CCRB farm can draw down 10 t of CO₂ per ha per year. This success is manifested by an increment in biomass accumulations and a decrease in the amount of dissolved inorganic carbon in the water column.

TAGS

general

m_cdr_focus: specific_mcdr | **m_cdr_method:** macro_algae_cultivation_and_sinking | **paper_type:** original_research | **paper_topic:** carbon_flux; mcd_r_method |

type_of_method_used: field_study | **adjacent_topic_to_m_cdr:** aquaculture;
blue_carbon | **adjacent_topic_to_fisheries:** aquaculture

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 7/31/2025

location

ocean_basin: pacific | **geopolitical_area:** asia | **habitat_type:** nearshore | **depth:**
not_applicable | **experiment_location:** field

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** macroalgae |
life_stage: not_applicable

treatment

exposure: not_applicable | **chemical_mineral_added:** not_applicable | **response_observed:** biological_effect_not_investigated

NOTES

summary_notes: Case study of carbon removal via kelp forests in southern Korea. The “Coastal CO₂ Removal Belt” is made up of natural and cultivated plant communities. Pilot study example cites brown alga (*Ecklonia*) drawing down 10 t of CO₂ per ha per year, increasing biomass accumulations and decreasing water column DIC.

mcd_r_relevance_notes: Pilot study of macroalgae cultivation and sequestration

fisheries_relevance_notes: Seaweed aquaculture

Biological impacts of enhanced alkalinity in *Carcinus maenas*

2013

Cripps, Gemma; Widdicombe, Stephen; Spicer, John I.; Findlay, Helen S.

Paper URL

Abstract: Further steps are needed to establish feasible alleviation strategies that are able to reduce the impacts of ocean acidification, whilst ensuring minimal biological side-effects in the process. Whilst there is a growing body of literature on the biological impacts of many other carbon dioxide reduction techniques, seemingly little is known about enhanced alkalinity. For this reason, we investigated the potential physiological impacts of using chemical sequestration as an alleviation strategy. In a controlled experiment, *Carcinus maenas* were acutely exposed to concentrations of Ca(OH)₂ that would be required to reverse the decline in ocean surface pH and return it to pre-industrial levels. Acute exposure significantly affected all individuals' acid-base balance resulting in slight respiratory alkalosis and hyperkalemia, which was strongest in mature females. Although the trigger for both of these responses is currently unclear, this

study has shown that alkalinity addition does alter acid-base balance in this comparatively robust crustacean species. (C) 2013 Elsevier Ltd. All rights reserved.

TAGS

general

m_cdr_focus: specific_mcdr | **m_cdr_method:** ocean_alkalinity_enhancement | **paper_type:** original_research | **paper_topic:** biological_ecological_impacts | **type_of_method_used:** lab_experiment | **adjacent_topic_to_m_cdr:** ocean_acidification | **adjacent_topic_to_fisheries:** higher_trophic

review_status

paper_review_status: complete_paper_review | **date_last_reviewed:** 5/13/2025

location

ocean_basin: not_applicable | **geopolitical_area:** europe | **habitat_type:** nearshore | **depth:** not_applicable | **experiment_location:** laboratory

species

species_common_name: Euro green crab | **species_scientific_name:** *Carcinus maenas* | **taxon:** crustacean | **life_stage:** not_applicable

treatment

exposure: chemical | **chemical_mineral_added:** other | **response_observed:** biological_effect_observed

NOTES

summary_notes: The study examines the effects of calcium hydroxide ($\text{Ca}(\text{OH})_2$) and calcium chloride (CaCl_2) on the physiology of the green crab (*Carcinus maenas*) and the alleviation of ocean acidification.

mcdr_relevance_notes: The findings from this study are pertinent to mCDR in the following ways: (1) Species-Specific Responses: The physiological stress observed in *C. maenas* suggests that other marine organisms might also experience adverse effects from increased alkalinity, highlighting the need for species-specific assessments. (2) Ecosystem Implications: As *C. maenas* is a common and ecologically significant species, its response to enhanced alkalinity could have broader implications for marine food webs and ecosystem dynamics. (3) Risk Assessment: The study underscores the importance of evaluating the potential biological impacts of mCDR strategies on marine organisms to ensure that such interventions do not inadvertently harm marine ecosystems.

fisheries_relevance_notes: NA

Impacts of ocean acidification and mitigative hydrated lime addition on Pacific oyster larvae: implications for shellfish aquaculture

2013

Duckham, Carolyn Susanne

[Paper URL](#)

Abstract: Anthropogenic CO₂ emissions are rapidly altering marine chemistry, significantly impacting calcifying organisms' ability to produce and maintain shells. In the Pacific Northwest, shellfish hatcheries have already observed mass die offs of larvae, potentially from ocean acidification. This study aims to elucidate if hydrated lime can be used at the hatchery level to chemically reverse ocean acidification and its negative impacts on larval stages of the Pacific oyster, *Crassostrea gigas*. For 24 days, larvae were exposed to four treatments: ambient - 390 ppm, ocean acidified - 1500 ppm, limed-ambient and limed-ocean acidified treatments to meet pre-industrial levels ~ 280 ppm. Hydrated lime significantly increased larval shell length after two weeks of development, but did not affect survival or percent abnormality between treatments. Investigations into mitigative lime use appear cost effective and feasible for shellfish hatcheries under future CO₂ scenarios, but requires more extensive research at the hatchery level, and for other species.

TAGS

general

m_cdr_focus: associated_fields | **m_cdr_method:** not_applicable | **paper_type:** original_research | **paper_topic:** biological_ecological_impacts; species_sensitivity | **type_of_method_used:** lab_experiment | **adjacent_topic_to_m_cdr:** aquaculture; ocean_acidification | **adjacent_topic_to_fisheries:** aquaculture; fisheries_practices; higher_trophic

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 8/20/2025

location

ocean_basin: not_applicable | **geopolitical_area:** canada | **habitat_type:** not_applicable | **depth:** not_applicable | **experiment_location:** laboratory

species

species_common_name: Pacific oyster | **species_scientific_name:** *Crassostrea gigas* | **taxon:** bivalves | **life_stage:** larvae

treatment

exposure: chemical | **chemical_mineral_added:** other | **response_observed:** biological_effect_observed

NOTES

summary_notes: Experimental study testing the use of hydrated lime for ocean acidification mitigation in Pacific oyster hatchery. 24-day treatment period with preindustrial pCO₂ target. Liming increased larval shell length after two weeks of development. No effect on survival or abnormalities.

mcd_r_relevance_notes: Liming in hatcheries (OAE) as an OA mitigation strategy

fisheries_relevance_notes: Mitigation of OA conditions via alkalinity additions; fisheries practices

Effects of CO₂ and iron availability on phytoplankton and eubacterial community compositions in the northwest subarctic Pacific

2013

Endo, H.; Yoshimura, T.; Kataoka, T.; Suzuki, K.

[Paper URL](#)

Abstract: On-deck CO₂-Fe-manipulated incubation experiments were conducted using surface seawater collected from the Western Subarctic Gyre of the NW Pacific in the summer of 2008 to elucidate the impacts of ocean acidification and Fe enrichment on the abundance and community composition of phytoplankton and eubacteria in the study area. During the incubation, excluding the initial period, the mean partial pressures of CO₂ in non-Fe-added bottles were 230, 419, 843, and 1124 atm, whereas those in Fe-added treatments were 152, 394, 791, and 1008 atm. Changes in the abundance and community composition of phytoplankton were estimated using HPLC pigment signatures with the program CHEMTAX and flow cytometry. A DGGE fingerprint technique targeting 16S rRNA gene fragments was also used to estimate changes in eubacterial phylotypes during incubation. The Fe addition induced diatom blooms, and subsequently stimulated the growth of heterotrophic bacteria such as *Roseobacter*, *Phaeobacter*, and *Alteromonas* in the post-bloom phase. In both the Fe-limited and Fe-replete treatments, concentrations of 19-hexanoyloxyfucoxanthin, a haptophyte marker, and the cell abundance of coccolithophores decreased at higher CO₂ levels (750 and 1000ppm), whereas diatoms exhibited little response to the changes in CO₂ availability. The abundances of *Synechococcus* and small eukaryotic phytoplankton (<10 μm) increased at the higher CO₂ levels. DGGE band positions revealed that *Methylobacterium* of Alphaproteobacteria occurred solely at lower CO₂ levels (180 and 380ppm) during the post-bloom phase. These results suggest that increases in CO₂ level could affect not only the community composition of phytoplankton but also that of eubacteria. As these microorganisms play critical roles in the biological carbon pump and microbial loop, our results indicate that the progression of ocean acidification can alter the biogeochemical processes in the study area.

TAGS

general

m_cdr_focus: associated_fields | **m_cdr_method:** not_applicable | **paper_type:** original_research | **paper_topic:** biological_ecological_impacts; nutrient_dynamics | **type_of_method_used:** lab_experiment | **adjacent_topic_to_m_cdr:** aquaculture; ocean_acidification | **adjacent_topic_to_fisheries:** aquaculture; biologic_carbon_pump; lower_trophic; nutrient_dynamics

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 8/21/2025

location

ocean_basin: pacific | **geopolitical_area:** not_applicable | **habitat_type:** open_ocean | **depth:** surface | **experiment_location:** field; laboratory

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** phytoplankton | **life_stage:** not_applicable

treatment

exposure: chemical | **chemical_mineral_added:** other | **response_observed:** biological_effect_observed

NOTES

summary_notes: Incubation experiments testing Fe enrichment effect on the abundance and community composition of phytoplankton. Fe enrichment caused diatom blooms and growth of heterotrophic bacteria.

mcd_r_relevance_notes: Nutrient fertilization of microalgae (not specifically for mCDR, but relevant bio response)

fisheries_relevance_notes: Response of phytoplankton to Fe additions

Enhanced chemical weathering as a geoengineering strategy to reduce atmospheric carbon dioxide, supply nutrients, and mitigate ocean acidification

2013

Hartmann, Jens; West, A. Joshua; Renforth, Phil; Köhler, Peter; De La Rocha, Christina L.; Wolf-Gladrow, Dieter A.; Dürr, Hans H.; Scheffran, Jürgen

[Paper URL](#)

Abstract: AbstractChemical weathering is an integral part of both the rock and carbon cycles and is being affected by changes in land use, particularly as a result of agricultural practices such as tilling, mineral fertilization, or liming to adjust soil pH. These human activities have already altered the terrestrial chemical cycles and land-ocean flux of major elements, although

the extent remains difficult to quantify. When deployed on a grand scale, Enhanced Weathering (a form of mineral fertilization), the application of finely ground minerals over the land surface, could be used to remove CO₂ from the atmosphere. The release of cations during the dissolution of such silicate minerals would convert dissolved CO₂ to bicarbonate, increasing the alkalinity and pH of natural waters. Some products of mineral dissolution would precipitate in soils or be taken up by ecosystems, but a significant portion would be transported to the coastal zone and the open ocean, where the increase in alkalinity would partially counteract “ocean acidification” associated with the current marked increase in atmospheric CO₂. Other elements released during this mineral dissolution, like Si, P, or K, could stimulate biological productivity, further helping to remove CO₂ from the atmosphere. On land, the terrestrial carbon pool would likely increase in response to Enhanced Weathering in areas where ecosystem growth rates are currently limited by one of the nutrients that would be released during mineral dissolution. In the ocean, the biological carbon pumps (which export organic matter and CaCO₃ to the deep ocean) may be altered by the resulting influx of nutrients and alkalinity to the ocean. This review merges current interdisciplinary knowledge about Enhanced Weathering, the processes involved, and the applicability as well as some of the consequences and risks of applying the method.

TAGS

general

m_cdr_focus: specific_mcdr | **m_cdr_method:** ocean_alkalinity_enhancement | **paper_type:** review | **paper_topic:** biological_ecological_impacts; carbon_flux; environmental_impacts; mcdr_method; nutrient_dynamics | **type_of_method_used:** not_applicable | **adjacent_topic_to_m_cdr:** ocean_acidification | **adjacent_topic_to_fisheries:** nutrient_dynamics

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 5/20/2025

location

ocean_basin: global | **geopolitical_area:** not_applicable | **habitat_type:** estuary; nearshore; terrestrial_watershed | **depth:** surface | **experiment_location:** not_applicable

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** not_applicable | **life_stage:** not_applicable

treatment

exposure: not_applicable | **chemical_mineral_added:** not_applicable | **response_observed:** biological_effect_not_investigated

NOTES

summary_notes: Review of current understanding of enhanced weathering and the impacts and benefits of adding minerals to land to stimulate production within the ocean. Talks about the nutrient fluxes as well and potential to mitigate OA

mcd_r_relevance_notes: Relevant to OAE in the ocean and enhanced weathering

fisheries_relevance_notes: Talks about nutrient supply, which impacts growth of biology

Assessing the potential of calcium-based artificial ocean alkalization to mitigate rising atmospheric CO₂ and ocean acidification: Modeling mitigation potential of AOA

2013

Ilyina, Tatiana; Wolf-Gladrow, Dieter; Munhoven, Guy; Heinze, Christoph

[Paper URL](#)

Abstract: Enhancement of ocean alkalinity using calcium compounds, e.g., lime has been proposed to mitigate further increase of atmospheric CO₂ and ocean acidification due to anthropogenic CO₂ emissions. Using a global model, we show that such alkalization has the potential to preserve pH and the saturation state of carbonate minerals at close to today's values. Effects of alkalization persist after termination: Atmospheric CO₂ and pH do not return to unmitigated levels. Only scenarios in which large amounts of alkalinity (i.e., in a ratio of 2:1 with respect to emitted CO₂) are added over large ocean areas can boost oceanic CO₂ uptake sufficiently to avoid further ocean acidification on the global scale, thereby elevating some key biogeochemical parameters, e.g., pH significantly above preindustrial levels. Smaller-scale alkalization could counteract ocean acidification on a subregional or even local scale, e.g., in upwelling systems. The decrease of atmospheric CO₂ would then be a small side effect.

TAGS

general

m_cdr_focus: specific_mcd_r | **m_cdr_method:** ocean_alkalinity_enhancement | **paper_type:** original_research | **paper_topic:** carbon_flux; environmental_impacts; mcd_r_method | **type_of_method_used:** modeling | **adjacent_topic_to_m_cdr:** ocean_acidification | **adjacent_topic_to_fisheries:** not_applicable

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 5/22/2025

location

ocean_basin: global | **geopolitical_area:** not_applicable | **habitat_type:** not_applicable | **depth:** not_applicable | **experiment_location:** not_applicable

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** not_applicable
| **life_stage:** not_applicable

treatment

exposure: not_applicable | **chemical_mineral_added:** not_applicable | **response_observed:** biological_effect_not_investigated

NOTES

summary_notes: Global modeling study on OAE and benefits of calcium-based techniques on OA and CO₂ uptake. Find that alkalization and saturation state can be preserved with OAE additions. Large scale deployments can remove CO₂. Smaller scales can be used to mitigate OA

mcd_r_relevance_notes: Modeling study on OAE to determine if it is feasible to remove CO₂ and possibly mitigate OA

fisheries_relevance_notes: Does not mention fisheries but mitigating OA would have positive impacts

Variation of phytoplankton and zooplankton communities in a sea area, with the building of an artificial upwelling structure

2013

Jeong, Yeon Kyu; Lee, Ha Neul; Park, Chan-Il; Kim, Dong Sun; Kim, Mu Chan

[Paper URL](#)

Abstract: In this study, we investigated changes in the standing crop, species composition, and ecological index of phytoplankton and zooplankton caused by the installation of an artificial upwelling structure in the sea area. The standing crop of phytoplankton and zooplankton in 2009, when the artificial upwelling structure was completed, increased 50-fold and 2.3-fold, respectively, from 2005 when the artificial upwelling structure was installed. During the investigation period, however, the correlation between the standing crop of zooplankton and the standing crop of phytoplankton did not show significant difference (SPSS, $p > 0.05$). Phytoplankton and zooplankton species showed a 2.47-fold increase in a total of 94 species and a 1.68-fold increase in a total of 27 species, respectively, from 2006 to 2009. During the study period, the dominant species of phytoplankton were *Chaetoceros didymus* (16.5%), *Chaetoceros pseudocrinitus* (8.5%), *Copelatus anastomosans* (7.6%), and *Skeletonema costatum* (6.0%), and the dominant species of zooplankton were *Paracalanus parvus* (21.6%), *Evadne tergestina* (6.6%), *Paracalanus aculeatus* (6.1%), *Lucifer intermedius* (5.7%), and *Sinocalanus tenellus* (5.4%), and copepods accounted for the majority of zooplankton. Phytoplankton species diversity index and richness index increased significantly from 1.68 to 2.45 and from 1.61 to 3.65, respectively, in the period from 2006 to 2009, but dominant species index decreased from 0.55

to 0.40 in the same period. It was known that bigger artificial upwelling structures caused a gradually rising upwelling effect and that in the study area, from 2006 to 2009, the ecosystem was changing to a more stable one where a variety of species habitat gradually.

TAGS

general

m_cdr_focus: specific_mcdr | **m_cdr_method:** artificial_upwelling | **paper_type:** original_research | **paper_topic:** biological_ecological_impacts; mcdr_method | **type_of_method_used:** field_study | **adjacent_topic_to_m_cdr:** not_applicable | **adjacent_topic_to_fisheries:** aquaculture; lower_trophic

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 9/5/2025

location

ocean_basin: pacific | **geopolitical_area:** asia | **habitat_type:** open_ocean | **depth:** mid_depth; surface | **experiment_location:** field

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** phytoplankton; zooplankton | **life_stage:** not_applicable

treatment

exposure: deep_water | **chemical_mineral_added:** nutrients | **response_observed:** biological_effect_observed

NOTES

summary_notes: Field study assessing the impact of artificial upwelling on standing crop, species composition, diversity, and richness of phytoplankton and zooplankton communities.

mcdr_relevance_notes: Pilot of artificial upwelling system

fisheries_relevance_notes: Biological and ecosystem response to artificial upwelling over multiple years

Responses of marine benthic microalgae to elevated CO₂

2013

Johnson, V. R.; Brownlee, C.; Rickaby, R. E. M.; Graziano, M.; Milazzo, M.; Hall-Spencer, J. M.

[Paper URL](#)

Abstract: Increasing anthropogenic CO₂ emissions to the atmosphere are causing a rise in pCO₂ concentrations in the ocean surface and lowering pH. To predict the effects of these

changes, we need to improve our understanding of the responses of marine primary producers since these drive biogeochemical cycles and profoundly affect the structure and function of benthic habitats. The effects of increasing CO₂ levels on the colonisation of artificial substrata by microalgal assemblages (periphyton) were examined across a CO₂ gradient off the volcanic island of Vulcano (NE Sicily). We show that periphyton communities altered significantly as CO₂ concentrations increased. CO₂ enrichment caused significant increases in chlorophyll a concentrations and in diatom abundance although we did not detect any changes in cyanobacteria. SEM analysis revealed major shifts in diatom assemblage composition as CO₂ levels increased. The responses of benthic microalgae to rising anthropogenic CO₂ emissions are likely to have significant ecological ramifications for coastal systems.

TAGS

general

m_cdr_focus: associated_fields | **m_cdr_method:** not_applicable | **paper_type:** original_research | **paper_topic:** biological_ecological_impacts; environmental_impacts; species_sensitivity | **type_of_method_used:** field_study | **adjacent_topic_to_m_cdr:** ocean_acidification | **adjacent_topic_to_fisheries:** lower_trophic

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 9/5/2025

location

ocean_basin: mediterranean | **geopolitical_area:** europe | **habitat_type:** nearshore | **depth:** benthic | **experiment_location:** field

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** phytoplankton | **life_stage:** not_applicable

treatment

exposure: natural_exposure | **chemical_mineral_added:** not_applicable | **response_observed:** biological_effect_observed

NOTES

summary_notes: Experiments testing the impact of high CO₂ on benthic microalgal assemblages. Communities were significantly altered as CO₂ increased.

mcd_r_relevance_notes: Benthic habitats may be exposed to high CO₂ conditions with some mCDR approaches

fisheries_relevance_notes: Structure and functioning of benthic habitats

The changes in marine environment and biological community in sea areas around an artificial upwelling structure during the summer

2013

Kim, Dong-Sun; Kim, Mu Chan

[Paper URL](#)

Abstract: In order to examine the changes in a biological community in the South Sea of Korea that is caused by the installation of an artificial upwelling structure during the summer season when the forming of the stratification is the strongest, the researchers evaluate the water's temperature and nutrients, the standing crops and species diversity of phytoplankton and zooplankton, and the benthic organisms. The forming depth of Maximum Thermocline of Water Column (MTWC) is elevated after installing an artificial upwelling structure. Also, instead of showing a clear stratification where the concentration of nutrients increases as the depth of the sea increases, the distribution of nutrients after installing an artificial upwelling structure has changed, showing a lower degree of stratification by causing the mixing of the water mass. Compared to 2005, when the artificial upwelling structure was first introduced, the standing crop of phytoplankton and zooplankton has increased by 58.2 and 30.0 times in 2008, thus showing a tendency to increase as the size of the artificial upwelling structure becomes bigger. Also, the species diversity index of phytoplankton and zooplankton has increased and the size of the artificial upwelling structure has become greater. When the environment underwater around the artificial upwelling structure was examined in 2008, the underwater environment, arthropods such as *Balanus albicostatus*, mollusks such as *Patinopecten yessoensis* and *Batillus cornutus*, echinoderms such as *Asterina pectinifera* and *Comanthus japonica*, chordates such as *Halocynthia roretzi*, and fishes such as *Sebastes inermis* and *Oplegnathus fasciatus* were found, thus forming a benthic environment.

TAGS

general

m_cdr_focus: associated_fields | **m_cdr_method:** not_applicable | **paper_type:** original_research | **paper_topic:** biological_ecological_impacts; environmental_impacts; nutrient_dynamics | **type_of_method_used:** field_study | **adjacent_topic_to_m_cdr:** not_applicable | **adjacent_topic_to_fisheries:** lower_trophic; nutrient_dynamics

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 9/8/2025

location

ocean_basin: pacific | **geopolitical_area:** asia | **habitat_type:** not_applicable | **depth:** benthic; mid_depth; surface | **experiment_location:** field

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** echinoderm; fish; other; phytoplankton; zooplankton | **life_stage:** not_applicable

treatment

exposure: deep_water | **chemical_mineral_added:** nutrients | **response_observed:** biological_effect_observed

NOTES

summary_notes: Study of the impact of an artificial upwelling structure on temperature, nutrients, phytoplankton, zooplankton, and benthic organisms.

mcd_r_relevance_notes: Biological and environmental impacts of artificial upwelling

fisheries_relevance_notes: Nutrient dynamics and lower trophic responses

Dissolved inorganic carbon and alkalinity fluxes from coastal marine sediments: model estimates for different shelf environments and sensitivity to global change

2013

Krumins, V.; Gehlen, M.; Arndt, S.; Van Cappellen, P.; Regnier, P.

[Paper URL](#)

Abstract: We present a one-dimensional reactive transport model to estimate benthic fluxes of dissolved inorganic carbon (DIC) and alkalinity (AT) from coastal marine sediments. The model incorporates the transport processes of sediment accumulation, molecular diffusion, bioturbation and bioirrigation, while the reactions included are the redox pathways of organic carbon oxidation, re-oxidation of reduced nitrogen, iron and sulfur compounds, pore water acid-base equilibria, and dissolution of particulate inorganic carbon (calcite, aragonite, and Mg-calcite). The coastal zone is divided into four environmental units with different particulate inorganic carbon (PIC) and particulate organic carbon (POC) fluxes: reefs, banks and bays, carbonate shelves and non-carbonate shelves. Model results are analyzed separately for each environment and then scaled up to the whole coastal ocean. The model-derived estimate for the present-day global coastal benthic DIC efflux is 126 Tmol yr⁻¹, based on a global coastal reactive POC depositional flux of 117 Tmol yr⁻¹. The POC decomposition leads to a carbonate dissolution from shallow marine sediments of 7 Tmol yr⁻¹ (on the order of 0.1 Pg C yr⁻¹). Assuming complete re-oxidation of aqueous sulfide released from sediments, the effective net flux of alkalinity to the water column is 29 Teq. yr⁻¹, primarily from PIC dissolution (46%) and ammonification (33%). Because our POC depositional flux falls in the high range of global values given in the literature, the reported DIC and alkalinity fluxes should be viewed as upper-bound estimates. Increasing coastal seawater DIC to what might be expected in year 2100 due to the uptake of anthropogenic CO₂ increases PIC dissolution by 2.3 Tmol yr⁻¹ and alkalinity efflux by 4.8 Teq. yr⁻¹. Our reactive transport modeling approach not only yields

global estimates of benthic DIC, alkalinity and nutrient fluxes under variable scenarios of ocean productivity and chemistry, but also provides insights into the underlying processes.

TAGS

general

m_cdr_focus: associated_fields | **m_cdr_method:** not_applicable | **paper_type:** original_research | **paper_topic:** carbon_flux | **type_of_method_used:** modeling | **adjacent_topic_to_m_cdr:** not_applicable | **adjacent_topic_to_fisheries:** not_applicable

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 5/23/2025

location

ocean_basin: not_applicable | **geopolitical_area:** not_applicable | **habitat_type:** nearshore | **depth:** not_applicable | **experiment_location:** not_applicable

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** not_applicable | **life_stage:** not_applicable

treatment

exposure: not_applicable | **chemical_mineral_added:** not_applicable | **response_observed:** biological_effect_not_investigated

NOTES

summary_notes: Reactive transport modeling of benthic DIC and alkalinity fluxes from coastal sediments. Fluxes are estimated separately for reefs, banks and bays, carbonate shelves and non-carbonate shelves, then scaled to the whole coastal ocean. Reported DIC and alkalinity fluxes are upper-bound estimates.

mcdr_relevance_notes: Estimates baseline coastal DIC and alkalinity fluxes, and projects potential alkalinity flux for increased anthropogenic CO₂ uptake by 2100.

fisheries_relevance_notes: Not relevant to fisheries.

Temporal and Spatial Variations and Species Composition of Ichthyoplanktons in a Sea Area, with the Construction of Artificial Upwelling Structure

2013

Kwak, Seok-Nam; Huh, Sung-Hoi; Kim, Ha-Won

[Paper URL](#)

Abstract: 2012 6 13, 9 21 25 .
 4,849 ind./\$1,000m³, 641 ind./\$1,000m³ . (Engraulis japonicus)
 57.5 % . (Unidentified) I, II, III, IV . (Pictiblennius yatabei) ,
 (Repomucenus sp.A, sp.B), (Scomber japonicus), (Engraulis japonicus), (Konosirus
 punctatus) . (Unidentified) I, II . , .
 7 , 8 . 2, 3 . 4

. Temporal and spatial variations and species composition of ichthyoplanktons were investigated by Bongo net in a sea area with the construction of artificial upwelling structure at 13, June, 21 September, and 23 September in 2012. A total of fish eggs was 4,849 ind./\$1,000m³, and fish larvae was 641 ind./\$1,000m³ in the study area. The common fish eggs was Engraulis japonicus which was composed of 57.5%, and unidentified I, II, III and IV were occurred with decreased number of individuals order. For fish larvae, Pictiblennius yatabei, Repomucenus sp.A, sp.B and Scomber japonicus were dominated, however, other species were Engraulis japonicus, Konosirus punctatus, and unidentified I, II. Temporal variations of fish eggs and larvae were remarkable differed. Higher numbers of fish eggs and larvae were July 2012, whereas those were lower at August 2012. Spatial variations of those were also significantly differed with higher numbers were at station 2 and 3 during study periods. The total number of fish eggs and larvae was about 4 fold against numbers of those in the compared area far from study area. These results indicated that higher numbers of fish eggs and larvae were due to influx much more nutrients and food organisms with artificial upwelling.

TAGS

general

m_cdr_focus: associated_fields | **m_cdr_method:** not_applicable | **paper_type:** original_research | **paper_topic:** biological_ecological_impacts; mcdr_method; nutrient_dynamics | **type_of_method_used:** field_study | **adjacent_topic_to_m_cdr:** aquaculture | **adjacent_topic_to_fisheries:** aquaculture; higher_trophic; nutrient_dynamics

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 9/9/2025

location

ocean_basin: pacific | **geopolitical_area:** asia | **habitat_type:** not_applicable | **depth:** surface | **experiment_location:** field

species

species_common_name: NA | **species_scientific_name:** Konosirus punctatus; Engraulis japonicus; Pictiblennius yatabei; Repomucenus sp.A, sp.B; Scomber japonicus | **taxon:** fish | **life_stage:** egg; larvae

treatment

exposure: deep_water | **chemical_mineral_added:** nutrients | **response_observed:** biological_effect_observed

NOTES

summary_notes: Field study assessed the abundance and species composition of fish eggs and larvae around an artificial upwelling structure. Results indicated a higher number of fish eggs and larvae could be due to the upwelling of nutrients.

mcd_r_relevance_notes: Ecological impact

fisheries_relevance_notes: Fish egg and larvae abundance and species composition at site of artificial upwelling

Geoengineering impact of open ocean dissolution of olivine on atmospheric CO₂, surface ocean pH and marine biology

2013

Köhler, Peter; Abrams, Jesse F; Völker, Christoph; Hauck, Judith; Wolf-Gladrow, Dieter A

[Paper URL](#)

Abstract: Ongoing global warming induced by anthropogenic emissions has opened the debate as to whether geoengineering is a ‘quick fix’ option. Here we analyse the intended and unintended effects of one specific geoengineering approach, which is enhanced weathering via the open ocean dissolution of the silicate-containing mineral olivine. This approach would not only reduce atmospheric CO₂ and oppose surface ocean acidification, but would also impact on marine biology. If dissolved in the surface ocean, olivine sequesters 0.28 g carbon per g of olivine dissolved, similar to land-based enhanced weathering. Silicic acid input, a byproduct of the olivine dissolution, alters marine biology because silicate is in certain areas the limiting nutrient for diatoms. As a consequence, our model predicts a shift in phytoplankton species composition towards diatoms, altering the biological carbon pumps. Enhanced olivine dissolution, both on land and in the ocean, therefore needs to be considered as ocean fertilization. From dissolution kinetics we calculate that only olivine particles with a grain size of the order of 1 μ m sink slowly enough to enable a nearly complete dissolution. The energy consumption for grinding to this small size might reduce the carbon sequestration efficiency by 30%.

TAGS

general

m_cdr_focus: specific_mcd_r | **m_cdr_method:** ocean_alkalinity_enhancement | **paper_type:** original_research | **paper_topic:** biological_ecological_impacts; carbon_flux; dissolution; mcd_r_method | **type_of_method_used:** modeling | **adjacent_topic_to_m_cdr:** not_applicable | **adjacent_topic_to_fisheries:** biologic_carbon_pump; lower_trophic

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 9/10/2025

location

ocean_basin: global | **geopolitical_area:** not_applicable | **habitat_type:** not_applicable | **depth:** not_applicable | **experiment_location:** not_applicable

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** phytoplankton | **life_stage:** not_applicable

treatment

exposure: not_applicable | **chemical_mineral_added:** not_applicable | **response_observed:** biological_effect_not_investigated

NOTES

summary_notes: Model study assessing the effectiveness of enhanced weathering of olivine for ocean alkalinity enhancement. Model predicts shifting phytoplankton species composition towards diatoms. Also considers the energy cost of grinding olivine to appropriate grain size.

mcd_r_relevance_notes: Model of OAE via olivine dissolution

fisheries_relevance_notes: Biological carbon pump, species composition

A systematic review of the effectiveness of liming to mitigate impacts of river acidification on fish and macro-invertebrates

2013

Mant, Rebecca C.; Jones, David L.; Reynolds, Brian; Ormerod, Steve J.; Pullin, Andrew S.

[Paper URL](#)

Abstract: The addition of calcium carbonate to catchments or watercourses – liming – has been used widely to mitigate freshwater acidification but the abatement of acidifying emissions has led to questions about its effectiveness and necessity. We conducted a systematic review and meta-analysis of the impact of liming streams and rivers on two key groups of river organisms: fish and invertebrates. On average, liming increased the abundance and richness of acid-sensitive invertebrates and increased overall fish abundance, but benefits were variable and not guaranteed in all rivers. Where B-A-C-I designs (before-after-control-impact) were used to reduce bias, there was evidence that liming decreased overall invertebrate abundance. This systematic review indicates that liming has the potential to mitigate the symptoms of acidification in some instances, but effects are mixed. Future studies should use robust designs to isolate recovery due to liming from decreasing acid deposition, and assess factors affecting liming outcomes.

TAGS

general

m_cdr_focus: associated_fields | **m_cdr_method:** not_applicable | **paper_type:** review | **paper_topic:** biological_ecological_impacts; nutrient_dynamics; species_sensitivity; thresholds | **type_of_method_used:** not_applicable | **adjacent_topic_to_m_cdr:** freshwater_environments | **adjacent_topic_to_fisheries:** aquaculture; higher_trophic

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 9/10/2025

location

ocean_basin: not_applicable | **geopolitical_area:** not_applicable | **habitat_type:** terrestrial_watershed | **depth:** not_applicable | **experiment_location:** not_applicable

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** fish; other | **life_stage:** not_applicable

treatment

exposure: not_applicable | **chemical_mineral_added:** not_applicable | **response_observed:** biological_effect_not_investigated

NOTES

summary_notes: Paper review of freshwater liming in rivers and lakes to mitigate impacts of OA on fish and macro-invertebrates. Found that liming increased abundance and richness

mcd_r_relevance_notes: OAE

fisheries_relevance_notes: Fish and invertebrates in freshwater

Iron fertilization enhanced net community production but not downward particle flux during the Southern Ocean iron fertilization experiment LOHAFEX

2013

Martin, Patrick; Van Der Loeff, Michiel Rutgers; Cassar, Nicolas; Vandromme, Pieter; d'Ovidio, Francesco; Stemmann, Lars; Rengarajan, R.; Soares, Melena; González, Humberto E.; Ebersbach, Friederike; Lampitt, Richard S.; Sanders, Richard; Barnett, Bruce A.; Smetacek, Victor; Naqvi, S. Wajih A.

[Paper URL](#)

Abstract: Abstract A closed eddy core in the Subantarctic Atlantic Ocean was fertilized twice with two tons of iron (as FeSO_4), and the 300 km² fertilized patch was studied for 39 days to test whether fertilization enhances downward particle flux into the deep ocean. Chlorophyll a and primary productivity doubled after fertilization, and photosynthetic quantum yield (F_v/F_m) increased from 0.33 to 0.40. Silicic acid ($<2 \mu\text{mol L}^{-1}$) limited diatoms, which

contributed <10% of phytoplankton biomass. Copepods exerted high grazing pressure. This is the first study of particle flux out of an artificially fertilized bloom with very low diatom biomass. Net community production (NCP) inside the patch, estimated from $O_2:Ar$ ratios, averaged $21 \text{ mmol POC m}^{-2} \text{ d}^{-1}$, probably $\pm 20\%$. ^{234}Th profiles implied constant export of $\sim 6.3 \text{ mmol POC m}^{-2} \text{ d}^{-1}$ in the patch, similar to unfertilized waters. The difference between NCP and ^{234}Th -derived export partly accumulated in the mixed layer and was partly remineralized between the mixed layer and 100 m. Neutrally buoyant sediment traps at 200 and 450 m inside and outside the patch caught mostly $< 1.1 \text{ mmol POC m}^{-2} \text{ d}^{-1}$, predominantly of fecal origin; flux did not increase upon fertilization. Our data thus indicate intense flux attenuation between 100 and 200 m, and probably between the mixed layer and 100 m. We attribute the lack of fertilization-induced export to silicon limitation of diatoms and reprocessing of sinking particles by detritus feeders. Our data are consistent with the view that nitrate-rich but silicate-deficient waters are not poised for enhanced particle export upon iron addition.

Key Points Primary productivity and net community production increased upon fertilization Due to silicon limitation, a bloom of small flagellates, not diatoms, developed Particle flux did not increase and was similar to fluxes in unfertilized waters

TAGS

general

m_cdr_focus: specific_mcdr | **m_cdr_method:** ocean_nutrient_fertilization | **paper_type:** original_research | **paper_topic:** biological_ecological_impacts; environmental_impacts; mcd_r_pilot; species_sensitivity | **type_of_method_used:** field_study | **adjacent_topic_to_m_cdr:** not_applicable | **adjacent_topic_to_fisheries:** biologic_carbon_pump; nutrient_dynamics

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 5/2/2025

location

ocean_basin: atlantic | **geopolitical_area:** united_states | **habitat_type:** open_ocean | **depth:** deep_ocean; mid_depth; surface | **experiment_location:** field

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** phytoplankton | **life_stage:** not_applicable

treatment

exposure: chemical | **chemical_mineral_added:** iron_sulfate | **response_observed:** biological_effect_observed

NOTES

summary_notes: LOHAFEX experiment fertilized iron (FeSO_4) in the Subantarctic Atlantic Ocean and studied particle flux · The first study to show low export of diatoms, likely due to silicate limited waters – nitrite-rich but silicate-deficient waters do not enhance particle flux · NCP increased, but flux attenuation between mixed layer and 100m, with decreased

flux between 100 and 200-450m. Zooplankton activity under mixed lay may regulate the flux as well

mcd_r__relevance__notes: OIF experiment

fisheries__relevance__notes: Looks a phytoplankton bloom response with phytoplankton and zooplankton as well as vertical particle flux

Challenges for the Kasatoshi volcano hypothesis as the cause of a large return of sockeye salmon (*Oncorhynchus nerka*) to the Fraser River in 2010

2013

McKinnell, Skip

[Paper URL](#)

Abstract: In August 2008 the Kasatoshi volcano in the Aleutian archipelago erupted. Prevailing winds carried volcanic ash to the Gulf of Alaska, where its soluble iron dissolved and initiated a widespread phytoplankton bloom. Two years later, the abundance of sockeye salmon (*Oncorhynchus nerka*) returning to spawn in the Fraser River was larger than any observed since 1913. Kasatoshi's influence on growth and survival has been proposed as the ultimate cause of the abundant return. However, when relevant data are brought to bear on the hypothesis, it shows that: (1) survival of the abundant sockeye salmon cohort was unremarkable when compared with the historical record; (2) without an accompanying volcano, survival of the returns in 2011 was about the same as in 2010; (3) parental abundance that spawned the abundant return in 2010 was the sixth largest since 1948; (4) during their first summer at sea, sockeye salmon are not known to migrate in the offshore region where the anomalous chlorophyll bloom occurred; (5) an older cohort that was likely exposed to the chlorophyll bloom had the second lowest productivity on record when it returned in 2009; (6) immature sockeye salmon do not feed on diatoms, so any trophodynamic benefit derived from excess chlorophyll in mid-to late August 2008 would have to have found a rapid trophic pathway to sockeye salmon on the continental shelf; and (7) no other populations of sockeye salmon or other species of salmon that are known to rear in the same region had unexpectedly high returns.

TAGS

general

m_cdr_focus: specific_mcd_r | **m_cdr_method:** ocean_nutrient_fertilization | **paper_type:** original_research | **paper_topic:** biological_ecological_impacts; environmental_impacts; natural_analogue; species_sensitivity | **type_of_method_used:** observation | **adjacent_topic_to_m_cdr:** not_applicable | **adjacent_topic_to_fisheries:** higher_trophic; nutrient_dynamics

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 5/2/2025

location

ocean_basin: pacific | **geopolitical_area:** united_states | **habitat_type:** nearshore;
open_ocean | **depth:** surface | **experiment_location:** field

species

species_common_name: Sockeye salmon | **species_scientific_name:** Oncorhynchus
nerka | **taxon:** fish | **life_stage:** adult; juvenile

treatment

exposure: natural_exposure | **chemical_mineral_added:** dust | **response_observed:**
biological_effect_observed

NOTES

summary_notes: This study investigated the claim that the Kasatoshi volcanic eruption led to the increased abundance of sockeye salmon two years later – it presents many challenges to this idea including: o Survival of returns would be the same without a volcano o Sockeye salmon are not known to migrate to the area with the Chl-a bloom during first summer o Older cohort likely exposed to the bloom had second lowest productivity o Immature sockeye salmon do not feed on diatoms o No other populations of salmon species that rear in the area has unexpectedly high returns

mcd_r_relevance_notes: Look at natural fertilization of volcano

fisheries_relevance_notes: Argues that Pacific Salmon did not increase in stock due to the input of iron

A simplified CFD model for air-lift artificial upwelling

2013

Meng, Qicheng; Wang, Chizhong; Chen, Ying; Chen, Jiawang

[Paper URL](#)

Abstract: Air-lift artificial upwelling was simulated through a simplified computational fluid dynamics (CFD) model of two-phase flow in an upward pipe. This model is based on the momentum transfer between the two phases. An algebraic turbulence model was employed to enclose the equations. The volumetric flow rates of the liquid and the gas phases, the void fraction distribution, and the pressure and velocity profiles along the axial coordinate were calculated in the stationary state. Numerical tests were conducted to verify the model. The good agreement obtained between the computational results and the field test measurements with cross-shaped air injectors demonstrates the capability of this model for dispersed bubbly

flow conditions. Comparisons with the analytical theory were also made. The results indicated that the proposed engineering-oriented model is efficient for the optimization of the system.

TAGS

general

m_cdr_focus: specific_mcdr | **m_cdr_method:** artificial_upwelling | **paper_type:** original_research | **paper_topic:** mcdr_method; tech_engineering | **type_of_method_used:** modeling | **adjacent_topic_to_m_cdr:** not_applicable | **adjacent_topic_to_fisheries:** not_applicable

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 9/9/2025

location

ocean_basin: not_applicable | **geopolitical_area:** not_applicable | **habitat_type:** not_applicable | **depth:** not_applicable | **experiment_location:** not_applicable

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** not_applicable | **life_stage:** not_applicable

treatment

exposure: not_applicable | **chemical_mineral_added:** not_applicable | **response_observed:** biological_effect_not_investigated

NOTES

summary_notes: Looks at computational fluid dynamics of artificial upwelling

mcdr_relevance_notes: AU

fisheries_relevance_notes: does not mention

Assessing possible consequences of ocean liming on ocean pH, atmospheric CO₂ concentration and associated costs

2013

Paquay, François S.; Zeebe, Richard E.

[Paper URL](#)

Abstract: More than 380Pg of carbon have been released by human activities from fossil fuel combustion and cement manufacturing since the beginning of the Industrial Revolution. This carbon release has caused atmospheric CO₂ levels to increase by 100ppmv compared to the highest interglacial value of the last 800,000 years. Net uptake of carbon by the oceans has caused the surface ocean pH to drop by 0.1 pH units. Evaluating carbon sequestration options

is now timely. We use the carbon-cycle model LOSCAR to evaluate the effects of artificially enhancing ocean alkalinity from year 2020 to year 2400 on ocean pH and atmospheric pCO₂. Ten different carbon emission scenarios with scaled alkalinity input are simulated. Results show that for ocean pH to be maintained above 8.0, on the order of 2–10×10¹⁴ moles of alkalinity/year for C emissions of 1500–5000Pg C, respectively, is required. Atmospheric pCO₂ remains high (500–600ppmv) in nine of the carbon emission scenarios. Ocean alkalization, if ever implemented at such a large scale, could allow ocean pH to be stabilized but pCO₂ would not return to pre-industrial levels. Cost estimates of the ocean alkalization operation, using quicklime (CaO) was estimated to range from 0.5 to 2.8\$US trillion/year, depending on the target pH selected to avoid damage to marine organisms and ecosystems. For comparison, this cost is 0.8% and 4.6% of the 2011 global Gross Domestic Product for total anthropogenic carbon emissions of 1500 and 5000Pg C, respectively. The cost of 1ton of CO₂ sequestered ranges from 103 to 144\$US for total emissions of 1500–5000Pg C, respectively. Producing large amounts of quicklime (without carbon capture and storage) would cause substantial additional CO₂ release. This reduces the fossil fuel emissions allowed to maintain pH above 8.0 by a factor of 1.3–2.5 (for total emissions of 1000 and 5000Pg C, respectively).

TAGS

general

m_cdr_focus: specific_mcdr | **m_cdr_method:** ocean_alkalinity_enhancement | **paper_type:** original_research | **paper_topic:** air_sea_gas_exchange; carbon_flux; socio_economic; tech_engineering | **type_of_method_used:** modeling | **adjacent_topic_to_m_cdr:** not_applicable | **adjacent_topic_to_fisheries:** not_applicable

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 8/30/2025

location

ocean_basin: global | **geopolitical_area:** not_applicable | **habitat_type:** not_applicable | **depth:** not_applicable | **experiment_location:** not_applicable

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** not_applicable | **life_stage:** not_applicable

treatment

exposure: not_applicable | **chemical_mineral_added:** not_applicable | **response_observed:** biological_effect_not_investigated

NOTES

summary_notes: Looks at modelling scenarios for OAE. Considers future climate scenarios with higher pCO₂ and techno-economic analysis

mcdr_relevance_notes: Looks at OAE

fisheries_relevance_notes: Does nto mention

Iron fertilization and the structure of planktonic communities in high nutrient regions of the Southern Ocean

2013

Quéguiner, Bernard

[Paper URL](#)

Abstract: In this review article, plankton community structure observations are analyzed both for artificial iron fertilization experiments and also for experiments dedicated to the study of naturally iron-fertilized systems in the Atlantic, Indian and Pacific sectors of the Southern Ocean in the POOZ (Permanently Open Ocean Zone) and the PFZ (Polar Frontal Zone). Observations made in natural systems are combined with those from artificially perturbed systems, in order to evaluate the seasonal evolution of pelagic communities, taking into account controlling factors related to the life cycles and the ecophysiology of dominant organisms. The analysis considers several types of planktonic communities, including both autotrophs and heterotrophs. These communities are spatially segregated owing to different life strategies. A conceptual general scheme is proposed to account for these observations and their variability, regardless of experiment type. Diatoms can be separated into 2 groups: Group 1 has slightly silicified fast growing cells that are homogeneously distributed in the surface mixed layer, and Group 2 has strongly silicified slowly growing cells within discrete layers. During the growth season, Group 1 diatoms show a typical seasonal succession of dominant species, within time windows of development that are conditioned by physical factors (light and temperature) as well as endogenous specific rhythms (internal clock), and biomass accumulation is controlled by the availability of nutrients. Group 1 diatoms are not directly grazed by mesozooplankton which is fed by protozooplankton, linking the microbial food web to higher trophic levels. Instead, successive dominant species of Group 1 are degraded via bacterial activity at the end of their growth season. Organic detritus fragments feed protozooplankton and mesozooplankton. The effective silicon pump leads to the progressive disappearance of silicic acid in surface waters. In contrast, Group 2 is resistant to grazing due to its strong silicification, and its biomass accumulates continuously but relatively slowly throughout the productive period. Group 2 diatoms are concentrated at or near the seasonal pycnocline and thus benefit from upward nutrient fluxes by diapycnal mixing. The decrease in light and the deep convective mixing in the fall produce both light and nutrient limitation leading to a massive carbon export of Group 2 diatoms, a major annual event of the biological pump. This scheme describes the seasonal evolution of plankton communities in surface waters of the Southern Ocean. The scheme could probably be extended to ecosystems that are characterized by a seasonal bloom under influence of iron or other nutrients.

TAGS

general

m_cdr_focus: specific_mcdr | **m_cdr_method:** ocean_nutrient_fertilization | **paper_type:** review | **paper_topic:** biological_ecological_impacts; environmental_impacts; mcd_r_pilot; mcd_r_review; species_sensitivity | **type_of_method_used:** not_applicable | **adjacent_topic_to_m_cdr:** not_applicable | **adjacent_topic_to_fisheries:** lower_trophic

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 4/17/2025

location

ocean_basin: southern_ocean | **geopolitical_area:** not_applicable | **habitat_type:** open_ocean | **depth:** surface | **experiment_location:** not_applicable

species

species_common_name: diatoms | **species_scientific_name:** NA | **taxon:** phytoplankton | **life_stage:** not_applicable

treatment

exposure: not_applicable | **chemical_mineral_added:** not_applicable | **response_observed:** biological_effect_not_investigated

NOTES

summary_notes: · Review of artificial and natural iron fertilization experiments/observations and impact on plankton community structure · Planktonic communities across autotrophs and heterotrophs are spatially segregated due to different life strategies · Diatoms group 1 (slightly silicified fast growing) and group 2 (strongly silicified slow growing) are separated in growth and life cycles. Group 1 degraded by bacterial activity, group 2 export beyond the photic zone and contribute to the biological pump

mcd_r_relevance_notes: Review paper about iron fertilization

fisheries_relevance_notes: Looks at growth dynamics for species in the lower trophic levels

Direct electrolytic dissolution of silicate minerals for air CO₂ mitigation and carbon-negative H₂ production

2013

Rau, Greg H.; Carroll, Susan A.; Bourcier, William L.; Singleton, Michael J.; Smith, Megan M.; Aines, Roger D.

[Paper URL](#)

Abstract: We experimentally demonstrate the direct coupling of silicate mineral dissolution with saline water electrolysis and H₂ production to effect significant air CO₂ absorption, chemical conversion, and storage in solution. In particular, we observed as much as a 105-fold increase in OH[−] concentration (pH increase of up to 5.3 units) relative to experimental controls following the electrolysis of 0.25 M Na₂SO₄ solutions when the anode was encased in powdered silicate mineral, either wollastonite or an ultramafic mineral. After electrolysis, full equilibration of the alkalized solution with air led to a significant pH reduction and as much as a 45-fold increase in dissolved inorganic carbon concentration. This demonstrated significant spontaneous air CO₂ capture, chemical conversion, and storage as a bicarbonate, predominantly as NaHCO₃. The excess OH[−] initially formed in these experiments apparently resulted via neutralization of the anolyte acid, H₂SO₄, by reaction with the base mineral silicate at the anode, producing mineral sulfate and silica. This allowed the NaOH, normally generated at the cathode, to go unneutralized and to accumulate in the bulk electrolyte, ultimately reacting with atmospheric CO₂ to form dissolved bicarbonate. Using nongrid or nonpeak renewable electricity, optimized systems at large scale might allow relatively high-capacity, energy-efficient (<300 kJ/mol of CO₂ captured), and inexpensive (<\$100 per tonne of CO₂ mitigated) removal of excess air CO₂ with production of carbon-negative H₂. Furthermore, when added to the ocean, the produced hydroxide and/or (bi)carbonate could be useful in reducing sea-to-air CO₂ emissions and in neutralizing or offsetting the effects of ongoing ocean acidification.

TAGS

general

m_cdr_focus: specific_mcdr | **m_cdr_method:** ocean_alkalinity_enhancement | **paper_type:** original_research | **paper_topic:** mcdr_method | **type_of_method_used:** lab_experiment | **adjacent_topic_to_m_cdr:** ocean_acidification | **adjacent_topic_to_fisheries:** not_applicable

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 5/15/2025

location

ocean_basin: not_applicable | **geopolitical_area:** not_applicable | **habitat_type:** not_applicable | **depth:** not_applicable | **experiment_location:** laboratory

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** not_applicable | **life_stage:** not_applicable

treatment

exposure: not_applicable | **chemical_mineral_added:** not_applicable | **response_observed:** biological_effect_not_investigated

NOTES

summary_notes: Experimental study combining silicate mineral dissolution and saline water electrolysis and hydrogen production. Anode encased in powdered silicate mineral was more effective at generating alkalinity than control experiments. Equilibration of the alkalized solution with air reduced pH and increased DIC significantly (captured CO₂). Study estimated that large scale systems could capture significant CO₂ while producing carbon-negative hydrogen.

mcd_r_relevance_notes: Electrochemical approach to ocean alkalinity enhancement with co-production of H₂.

fisheries_relevance_notes: Not relevant to fisheries.

Engineering challenges of ocean liming

2013

Renforth, P.; Jenkins, B. G.; Kruger, T.

[Paper URL](#)

Abstract: The relationship between the level of atmospheric CO₂ (carbon dioxide) and the impacts of climate change is uncertain, but a safe concentration may be surpassed this century. Therefore, it is necessary to develop technologies that can accelerate CO₂ removal from the atmosphere. This paper explores the engineering challenges of a technology that manipulates the carbonate system in seawater by the addition of calcium oxide powder (CaO; lime), resulting in a net sequestration of atmospheric CO₂ into the ocean (ocean liming; OL). Every tonne of CO₂ sequestered requires between 1.4 and 1.7 t of limestone to be crushed, calcined, and distributed. Approximately 1 t of CO₂ would be created from this activity, of which >80% is a high purity gas (pCO₂ > 98%) amenable to geological storage. It is estimated that the thermal and electrical energy requirements for OL would be 0.6–5.6 and 0.1–1.2 GJ tCO₂^{−1} captured respectively. A preliminary economic assessment suggests that OL could cost approximately US\$72–159 t^{−1} of CO₂. The additional CO₂ burden of OL makes it a poor alternative to point source mitigation. However, it may provide a means to mitigate some diffuse emissions and reduce atmospheric concentrations.

TAGS

general

m_cdr_focus: specific_mcd_r | **m_cdr_method:** ocean_alkalinity_enhancement | **paper_type:** original_research | **paper_topic:** carbon_flux; dissolution; mcd_r_method; socio_economic; tech_engineering | **type_of_method_used:** modeling | **adjacent_topic_to_m_cdr:** other | **adjacent_topic_to_fisheries:** not_applicable

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 8/18/2025

location

ocean_basin: not_applicable | **geopolitical_area:** not_applicable | **habitat_type:** not_applicable | **depth:** not_applicable | **experiment_location:** not_applicable

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** not_applicable | **life_stage:** not_applicable

treatment

exposure: not_applicable | **chemical_mineral_added:** not_applicable | **response_observed:** biological_effect_not_investigated

NOTES

summary_notes: Assesses potential for ocean liming to sequester carbon. Considered the method, efficiency, and economic costs

mcd_r_relevance_notes: Relevant to OAE

fisheries_relevance_notes: does not mention

Coupling Mineral Carbonation and Ocean Liming

2013

Renforth, P.; Kruger, T.

[Paper URL](#)

Abstract: The process by which basic/ultrabasic silicate minerals (e.g., olivine) are reacted with CO₂ to produce solid carbonate minerals (“mineral carbonation”) has been suggested as a method to sequester carbon dioxide from point sources into stable carbonate minerals. Alternatively, the addition of lime (produced from calcining carbonate minerals) to the surface ocean (“ocean liming”), which results in an increase in ocean pH and a draw-down of atmospheric CO₂ has been proposed as a “geoengineering” technology, which stores carbon as dissolved alkalinity in the surface ocean. Combining these approaches, in which the magnesium carbonate minerals produced from mineral carbonation are used as a feedstock for ocean liming (mineral carbonation-ocean liming; MC-OL), may reduce the limitations of individual technologies while maximizing the benefits. Approximately 1.9 metric tons of magnesium silicate (producing 0.7 ton of magnesium oxide) are required for every net ton of CO₂ sequestered. A total of 0.7 ton of CO₂ is produced from this activity, 70% of which is high-purity (>98%) from calcining and potentially amenable for geological storage. The technology can be conceptually viewed as an alternative to direct air capture and swaps ambient CO₂ for high-purity point source CO₂. MC-OL requires approximately 4.9 and 2.2 GJ of thermal and electrical energy ton⁻¹ of CO₂ sequestered. MC-OL has less demand for geological storage; only 0.5 ton of CO₂ needs to be injected for every ton of CO₂ removed from the atmosphere. However,

manipulation of ocean chemistry in this way potentially creates an additional environmental impact (localized elevated pH or co-dissolution of trace metals) and requires additional attention.

TAGS

general

m_cdr_focus: specific_mcdr | **m_cdr_method:** ocean_alkalinity_enhancement | **paper_type:** original_research | **paper_topic:** carbon_flux; dissolution; mcdr_method | **type_of_method_used:** modeling | **adjacent_topic_to_m_cdr:** fundamental_carbon_chem | **adjacent_topic_to_fisheries:** not_applicable

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 8/18/2025

location

ocean_basin: not_applicable | **geopolitical_area:** not_applicable | **habitat_type:** not_applicable | **depth:** not_applicable | **experiment_location:** not_applicable

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** not_applicable | **life_stage:** not_applicable

treatment

exposure: not_applicable | **chemical_mineral_added:** not_applicable | **response_observed:** biological_effect_not_investigated

NOTES

summary_notes: methodology for OAE using liming. Talks about efficiency and amount of liming needed. mentions a need to understand the environmental impacts

mcdr_relevance_notes: OAE methodology

fisheries_relevance_notes: DOes not mention

Microalgae: a promising tool for carbon sequestration

2013

Singh, Uday Bhan; Ahluwalia, A. S.

[Paper URL](#)

Abstract: Increasing trends in global warming already evident, the likelihood of further rise continuing, and their impacts give urgency to addressing carbon sequestration technologies more coherently and effectively. Carbon dioxide (CO₂) is responsible for over half the warming potential of all greenhouse gases (GHG), due to the dependence of world economies on

fossil fuels. The processes involving CO₂ capture and storage (CCS) are gaining attention as an alternative for reducing CO₂ concentration in the ambient air. However, these technologies are considered as short-term solutions, as there are still concerns about the environmental sustainability of these processes. A promising technology could be the biological capture of CO₂ using microalgae due to its unmatched advantages over higher plants and ocean fertilization. Microalgae are phototrophic microorganisms with simple nutritional requirements, and comprising the major primary producers on this planet. Specific pathways include autotrophic production via both open pond or closed photobioreactor (PBR) systems. Photosynthetic efficiency of microalgae ranged from 10–20 % in comparison with 1–2 % of most terrestrial plants. Some algal species, during their exponential growth, can double their biomass in periods as short as 3.5 hours. Moreover, advantage of being tolerant of high concentration of CO₂ (flue gas), low light intensity requirements, environmentally sustainable, and co-producing added value products put these as the favoured organisms. Advantages of microalgae in comparison with other sequestration methodologies are discussed, which includes the cultivation systems, the key process parameters, wastewater treatment, harvesting and the novel bio-products produced by microalgal biomass.

TAGS

general

m_cdr_focus: specific_mcdr | **m_cdr_method:** macro_algae_cultivation_and_sinking | **paper_type:** original_research | **paper_topic:** carbon_flux; mcd_r_method; species_sensitivity | **type_of_method_used:** modeling | **adjacent_topic_to_m_cdr:** aquaculture | **adjacent_topic_to_fisheries:** aquaculture

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 8/9/2025

location

ocean_basin: not_applicable | **geopolitical_area:** not_applicable | **habitat_type:** not_applicable | **depth:** not_applicable | **experiment_location:** not_applicable

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** macroalgae | **life_stage:** not_applicable

treatment

exposure: not_applicable | **chemical_mineral_added:** not_applicable | **response_observed:** biological_effect_not_investigated

NOTES

summary_notes: Present an overview of macroalgae as a possible carbon sequestration mechanism. Talk about growth potential and photobioreactors

mcd_r_relevance_notes: macroalgae

fisheries_relevance_notes: relevant to aquaculture

Altered Oceanic pH Impairs Mating Propensity in a Pipefish

2013

Sundin, Josefin; Rosenqvist, Gunilla; Berglund, Anders

[Paper URL](#)

Abstract: Abstract Anthropogenic disturbance is currently altering the environment of terrestrial as well as aquatic organisms. Those changes affect a variety of animal behaviours, which in turn may cause changes in species interactions, population dynamics and evolutionary processes. In marine ecosystems, nutrient enrichment may elevate pH, while it is reduced by carbon dioxide-induced ocean acidification. These two processes are not expected to balance one another but rather to affect the environment at different times and scales. We here show experimentally that an increase in water pH has a negative effect on mating propensity in the broad-nosed pipefish *Syngnathus typhle*, whereas lowered pH did not elicit the same detrimental effect. This study provides, to our knowledge, the first evidence that mating propensity is impaired by an increase in pH, suggesting that anthropogenic nutrient enrichment in aquatic ecosystems may change the processes of sexual selection and population dynamics solely on the basis of altered water pH.

TAGS

general

m_cdr_focus: associated_fields | **m_cdr_method:** not_applicable | **paper_type:** original_research | **paper_topic:** biological_ecological_impacts; species_sensitivity | **type_of_method_used:** lab_experiment | **adjacent_topic_to_m_cdr:** other | **adjacent_topic_to_fisheries:** higher_trophic

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 7/31/2025

location

ocean_basin: not_applicable | **geopolitical_area:** not_applicable | **habitat_type:** not_applicable | **depth:** not_applicable | **experiment_location:** laboratory

species

species_common_name: broad-nosed pipefish | **species_scientific_name:** *Syngnathus typhle* | **taxon:** fish | **life_stage:** adult

treatment

exposure: chemical | **chemical_mineral_added:** other | **response_observed:** biological_effect_observed

NOTES

summary_notes: Looks at animal behavior for *Syngnathus typhle* broad-nosed pipefish. Found mating behaviors inhibited at higher pH. did not see same impact with lower pH

mcd_r_relevance_notes: Relevant to increased pH and nutrient additions

fisheries_relevance_notes: looks at broad-nosed pipefish mating

Effects of low pCO₂ conditions on sea urchin larval size

2013

Suwa, Ryota; Nojiri, Yukihiro; Ono, Tsuneo; Shirayama, Yoshihisa

[Paper URL](#)

Abstract: Ocean acidification results from an increase in the concentrations of atmospheric carbon dioxide (CO₂) impacts on marine calcifying species, which is predicted to become more pronounced in the future. By the end of this century, atmospheric pCO₂ levels will have doubled relative to the pre-industrial levels of 280 ppm. However, the effects of pre-industrial pCO₂ levels on marine organisms remain largely unknown. In this study, we investigated the effects of pre-industrial pCO₂ conditions on the size of the pluteus larvae of sea urchins, which are known to be vulnerable to ocean acidification. The larval size of *Hemicentrotus pulcherrimus* significantly increased when reared at pre-industrial pCO₂ level relative to the present one, and the size of *Anthocidaris crassispina* larvae decreased as the pCO₂ levels increased from the pre-industrial level to the near future ones after 3 days' exposure. In this study, it is suggested that echinoid larvae responded to pre-industrial pCO₂ levels. Ocean acidification may be affecting some sensitive marine calcifiers even at the present pCO₂ level.

TAGS

general

m_cdr_focus: associated_fields | **m_cdr_method:** not_applicable | **paper_type:** original_research | **paper_topic:** biological_ecological_impacts; species_sensitivity; thresholds | **type_of_method_used:** lab_experiment | **adjacent_topic_to_m_cdr:** ocean_acidification | **adjacent_topic_to_fisheries:** lower_trophic

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 7/31/2025

location

ocean_basin: not_applicable | **geopolitical_area:** not_applicable | **habitat_type:** not_applicable | **depth:** not_applicable | **experiment_location:** not_applicable

species

species_common_name: Sea urchin | **species_scientific_name:** *Hemicentrotus pulcherrimus*; *Anthocidaris crassispina* | **taxon:** echinoderm | **life_stage:** larvae

treatment

exposure: chemical | **chemical_mineral_added:** other | **response_observed:** biological_effect_observed

NOTES

summary__notes: Looks at preindustrial pCO₂ levels when rearing sea urchin larva and find enhanced growth

mcd_r__relevance__notes: OAE and DOCCS

fisheries__relevance__notes: sea urchin larvae

Effect of increased pCO₂ on the planktonic metabolic balance during a mesocosm experiment in an Arctic fjord

2013

Tanaka, T.; Alliouane, S.; Bellerby, R. G. B.; Czerny, J.; de Kluijver, A.; Riebesell, U.; Schulz, K. G.; Silyakova, A.; Gattuso, J.-P.

[Paper URL](#)

Abstract: The effect of ocean acidification on the balance between gross community production (GCP) and community respiration (CR) (i.e., net community production, NCP) of plankton communities was investigated in summer 2010 in Kongsfjorden, west of Svalbard. Surface water, which was characterized by low concentrations of dissolved inorganic nutrients and chlorophyll a (a proxy of phytoplankton biomass), was enclosed in nine mesocosms and subjected to eight pCO₂ levels (two replicated controls and seven enhanced pCO₂ treatments) for one month. Nutrients were added to all mesocosms on day 13 of the experiment, and thereafter increase of chlorophyll a was provoked in all mesocosms. No clear trend in response to increasing pCO₂ was found in the daily values of NCP, CR, and GCP. For further analysis, these parameters were cumulated for the following three periods: phase 1 – end of CO₂ manipulation until nutrient addition (t₄ to t₁₃); phase 2 – nutrient addition until the second chlorophyll a minimum (t₁₄ to t₂₁); phase 3 – the second chlorophyll a minimum until the end of this study (t₂₂ to t₂₈). A significant response was detected as a decrease of NCP with increasing pCO₂ during phase 3. CR was relatively stable throughout the experiment in all mesocosms. As a result, the cumulative GCP significantly decreased with increasing pCO₂ during phase 3. After the nutrient addition, the ratios of cumulative NCP to cumulative consumption of NO₃ and PO₄ showed a significant decrease during phase 3 with increasing pCO₂. The results suggest that elevated pCO₂ influenced cumulative NCP and stoichiometric C and nutrient coupling of the plankton community in a high-latitude fjord only for a limited period. However provided that there were some differences or weak correlations between NCP data based on different methods in the same experiment, this conclusion should be taken with caution.

TAGS

general

m_cdr__focus: associated_fields | **m_cdr__method:** not_applicable | **paper__type:**

original_research | **paper_topic:** biological_ecological_impacts; nutrient_dynamics;
species_sensitivity | **type_of_method_used:** mesocosm | **adjacent_topic_to_m_cdr:**
ocean_acidification | **adjacent_topic_to_fisheries:** lower_trophic; nutrient_dynamics

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 7/31/2025

location

ocean_basin: arctic | **geopolitical_area:** europe | **habitat_type:** nearshore | **depth:**
not_applicable | **experiment_location:** mesocosm

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** phytoplankton
| **life_stage:** not_applicable

treatment

exposure: chemical | **chemical_mineral_added:** nutrients; other | **response_observed:**
biological_effect_observed

NOTES

summary_notes: Only looks at enhanced pCO₂, not lower pCO₂, yet looks at a wide range
so may be relevant to OAE/DOCCS. Looks at phytoplankton response in a mesocosm

mcd_r_relevance_notes: Relevant to OAE and DOCCS

fisheries_relevance_notes: lower trophic levels and nutrient additions

**Sensitivity of Antarctic phytoplankton species to ocean acidification: Growth,
carbon acquisition, and species interaction**

2013

Trimborn, Scarlett; Brenneis, Tina; Sweet, Elizabeth; Rost, Björn

[Paper URL](#)

Abstract: Despite the fact that ocean acidification is considered to be especially pronounced in the Southern Ocean, little is known about CO₂-dependent physiological processes and the interactions of Antarctic phytoplankton key species. We therefore studied the effects of CO₂ partial pressure (PCO₂) (16.2, 39.5, and 101.3 Pa) on growth and photosynthetic carbon acquisition in the bloom-forming species *Chaetoceros debilis*, *Pseudo-nitzschia subcurvata*, *Fragilariopsis kerguelensis*, and *Phaeocystis antarctica*. Using membrane-inlet mass spectrometry, photosynthetic O₂ evolution and inorganic carbon (C_i) fluxes were determined as a function of CO₂ concentration. Only the growth of *C. debilis* was enhanced under high PCO₂. Analysis of the carbon concentrating mechanism (CCM) revealed the operation of very efficient

CCMs (i.e., high C_i affinities) in all species, but there were species-specific differences in CO₂-dependent regulation of individual CCM components (i.e., CO₂ and uptake kinetics, carbonic anhydrase activities). Gross CO₂ uptake rates appear to increase with the cell surface area to volume ratios. Species competition experiments with *C. debilis* and *P. subcurvata* under different PCO₂ levels confirmed the CO₂-stimulated growth of *C. debilis* observed in monospecific incubations, also in the presence of *P. subcurvata*. Independent of PCO₂, high initial cell abundances of *P. subcurvata* led to reduced growth rates of *C. debilis*. For a better understanding of future changes in phytoplankton communities, CO₂-sensitive physiological processes need to be identified, but also species interactions must be taken into account because their interplay determines the success of a species.

TAGS

general

m_cdr_focus: associated_fields | **m_cdr_method:** not_applicable | **paper_type:** original_research | **paper_topic:** biological_ecological_impacts; species_sensitivity | **type_of_method_used:** lab_experiment | **adjacent_topic_to_m_cdr:** fundamental_carbon_chem; ocean_acidification | **adjacent_topic_to_fisheries:** lower_trophic

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 7/30/2025

location

ocean_basin: southern_ocean | **geopolitical_area:** antarctica | **habitat_type:** not_applicable | **depth:** not_applicable | **experiment_location:** laboratory

species

species_common_name: NA | **species_scientific_name:** *Chaetoceros debilis*; *Pseudonitzschia subcurvata*; *Fragilariopsis kerguelensis*; *Phaeocystis antarctica* | **taxon:** phytoplankton | **life_stage:** not_applicable

treatment

exposure: low_co2; other | **chemical_mineral_added:** other | **response_observed:** biological_effect_observed

NOTES

summary_notes: Looked at carbonate processes and physiological functions of antarctica bloom forming phytoplankton. Looked at different pCO₂ concentrations

mcdr_relevance_notes: DOCCS and OAE where pH changes directly

fisheries_relevance_notes: Looks at lower trophic levels function

Plankton response to nutrient enrichment is maximized at intermediate distances from fish farms

2013

Tsagaraki, Tatiana M.; Pitta, Paraskevi; Frangoulis, Constantin; Petihakis, George; Karakassis, Ioannis

[Paper URL](#)

Abstract: Plankton community composition changes in response to nutrient enrichment were examined at 2 sites in the eastern Mediterranean. Samples from bacteria up to mesozooplankton were taken upstream and downstream of finfish farms in the north Aegean (Lesvos, Greece) and on the southern coast of Cyprus. The pattern of change appeared to be similar, albeit of different magnitude, in both areas. In Cyprus, results showed an increase in bacteria and a decrease in diatom abundance close to the cages, response to effluents was limited, and consequent growth was controlled by microzooplankton. In Lesvos, results showed increased abundance of bacteria, nanoflagellates and mesozooplankton, a decrease in diatoms and an increase in larger dinoflagellates and ciliates downstream of the farm. A shift towards a more diverse microplankton community was observed, consistent with the intermediate disturbance hypothesis. The community consisted of larger-sized microzooplankton, and changes persisted throughout the downstream stations. Results showed that the plankton response to nutrient enrichment is more pronounced at intermediate distances from the farm. The community composition changes observed in Lesvos indicate that even in exposed sites with high current velocities there is an influence of farm cages. Changes manifest in terms of size for some groups and abundance for others. Although fish farms have been associated with degradation of the environment in which they are located, this was not apparent in the present study. Indeed, the shifts induced by farm effluents can help us identify drivers of change and assess community responses to perturbations in a dynamic environment.

TAGS

general

m_cdr_focus: associated_fields | **m_cdr_method:** not_applicable | **paper_type:** original_research | **paper_topic:** biological_ecological_impacts; environmental_impacts; nutrient_dynamics | **type_of_method_used:** field_study | **adjacent_topic_to_m_cdr:** other | **adjacent_topic_to_fisheries:** aquaculture; nutrient_dynamics

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 7/29/2025

location

ocean_basin: mediterranean | **geopolitical_area:** europe | **habitat_type:** not_applicable | **depth:** not_applicable | **experiment_location:** field

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** microbes | **life_stage:** not_applicable

treatment

exposure: chemical | **chemical_mineral_added:** nutrients | **response_observed:** biological_effect_observed

NOTES

summary_notes: Relevant to phytoplankton growth during nutrient enrichment. Looks at downstream effects of nutrient addition from fish farms

mcd_r_relevance_notes: Relevant to ONF

fisheries_relevance_notes: Mentions microzooplankton and bacteria

Effects of elevated pCO₂ and the effect of parent acclimation on development in the tropical Pacific sea urchin *Echinometra mathaei*

2013

Uthicke, S.; Soars, N.; Foo, S.; Byrne, M.

[Paper URL](#)

Abstract: Effects of acclimation to projected near-future ocean acidification (OA) conditions on physiology, reproduction and development were investigated in the tropical sea urchin *Echinometra mathaei*. Following 6 weeks in control or one of the three elevated pCO₂ (pHNIST 7.5–8.1; pCO₂ ~485–1,770 atm) conditions, adult urchins exhibited a slight decline of growth in low pH treatments and moderately reduced respiration at intermediate levels. At 7 weeks, gametes from adults were used to produce larvae that were reared in their respective parental treatments. To assess whether larvae from acclimated parents are more resilient to elevated pCO₂ than those not acclimated, larvae from control animals were also reared in the elevated pCO₂ treatments. There was no difference in female ‘spawnability’ and oocyte size between treatments, but male spawning ability was reduced in increased pCO₂ conditions. In elevated pCO₂ treatments, the percentage of normal larvae and larval size decreased in the progeny of control- and elevated pCO₂-acclimated parents, and arm asymmetry increased. Thus, acclimation of the parents did not make the progeny more resilient or sensitive to OA effects. Negative effects of increased pCO₂ on reproduction and development may impact on recruitment and population maintenance of this species.

TAGS

general

m_cdr_focus: associated_fields | **m_cdr_method:** not_applicable | **paper_type:** original_research | **paper_topic:** biological_ecological_impacts; species_sensitivity; thresholds

| **type_of_method_used:** lab_experiment | **adjacent_topic_to_m_cdr:** ocean_acidification | **adjacent_topic_to_fisheries:** lower_trophic

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 7/29/2025

location

ocean_basin: not_applicable | **geopolitical_area:** not_applicable | **habitat_type:** not_applicable | **depth:** not_applicable | **experiment_location:** laboratory

species

species_common_name: Sea urchin | **species_scientific_name:** Echinometra mathaei | **taxon:** echinoderm | **life_stage:** adult; juvenile; larvae

treatment

exposure: chemical | **chemical_mineral_added:** other | **response_observed:** biological_effect_observed

NOTES

summary_notes: Looked at larvae recruitment and growth of sea urchin Echinometra mathaei at varying pH up to 8.1. Looked at different pH levels

mcd_r_relevance_notes: Relevant to OAE and DOCCS for pH variance

fisheries_relevance_notes: Mentions echinoderm

Retrospective satellite ocean color analysis of purposeful and natural ocean iron fertilization

2013

Westberry, Toby K.; Behrenfeld, Michael J.; Milligan, Allen J.; Doney, Scott C.

[Paper URL](#)

Abstract: Significant effort has been invested in understanding the role of iron in marine ecosystems over the past few decades. What began as shipboard amendment experiments quickly grew into a succession of in situ, mesoscale ocean iron fertilization (OIF) experiments carried out in all three high nutrient low chlorophyll (HNLC) regions of the world ocean. Dedicated process studies have also looked at regions of the ocean that are seasonally exposed to iron-replete conditions as natural OIF experiments. However, one problem common to many OIF experiments is determination of biological response beyond the duration of the experiment (typically o 1 month). Satellite-derived products have been used to address this shortcoming with some success, but thus far, have been limited snapshots of a single parameter, chlorophyll. Here, we investigate phytoplankton responses to OIF in both purposeful and naturally iron enriched systems using estimates of chlorophyll (Chl), phytoplankton carbon

biomass (Cphyto), their ratio (Chl:Cphyto) and two fluorescence indices, fluorescence per unit chlorophyll (FLH:Chl) and the chlorophyll fluorescence efficiency (ff). These quantities allow partitioning of the biological response to OIF into that due to changes in biomass and that due to phytoplankton physiology. We find that relative increases in Chl (~ 10–20x) following OIF far exceed increases in Cphyto (~ 4–5x), suggesting that a significant fraction of the observed Chl increase is associated with physiological adjustment to increased growth rates, photoacclimation, and floristic shifts in the phytoplankton community. Further, a consistent pattern of decreased satellite fluorescence efficiency (FLH:Chl or ff) following OIF is observed that is in agreement with current understanding of phytoplankton physiological responses to relief from iron stress. The current study extends our ability to retrieve phytoplankton physiology from space-based sensors, strengthens the link between satellite fluorescence and iron availability, and shows that satellite ocean color analyses provide a unique tool for monitoring OIF experiments.

TAGS

general

m_cdr_focus: specific_mcdr | **m_cdr_method:** ocean_nutrient_fertilization | **paper_type:** original_research | **paper_topic:** biological_ecological_impacts; environmental_impacts; natural_analogue | **type_of_method_used:** observation | **adjacent_topic_to_m_cdr:** not_applicable | **adjacent_topic_to_fisheries:** lower_trophic; nutrient_dynamics

review_status

paper_review_status: abstract_reviewed | **date_last_reviewed:** 5/7/2025

location

ocean_basin: global | **geopolitical_area:** not_applicable | **habitat_type:** open_ocean | **depth:** surface | **experiment_location:** field

species

species_common_name: NA | **species_scientific_name:** NA | **taxon:** phytoplankton | **life_stage:** not_applicable

treatment

exposure: natural_exposure | **chemical_mineral_added:** dust | **response_observed:** biological_effect_observed

NOTES

summary_notes: Used satellite imagery to look at changes in phytoplankton biomass/growth after natural OIF for longer periods of time. Sees increased Chl but not necessarily increases in phytoplankton biomass, suggesting increased growth rate.

mcdr_relevance_notes: Related to OIF

fisheries_relevance_notes: Related to lower trophic levels and phtoplankton